

Question

Element **X** is one of the essential non-metallic elements for life. Its simple substance reacts with Cl_2 to obtain a compound **A** with a structure similar to H_2O_2 . **A** hydrolyzes and disproportionates into the simple substance and **B**. **A** reacts with O_3 to obtain **B** and **C**. The simple substance reacts with ClF to obtain **D**. The oxidation states of **X** in **B**, **C**, and **D** are all the same, all are binary compounds, and **B** mainly exhibits oxidizing properties in chemical reactions. **D** and the highest-valent oxide **E** of **X** react in a 1:1 ratio to obtain **F** and **G** in a 1:1 ratio. **F** and **G** are both ternary compounds, where **F** can also be obtained by reacting the hydrate **H** of **G** with fluorosulfuric acid. **H** is a strong acid.

Please select the correct options from the following statements.

- A.** From the information in the question, we know that **X** is a non-metallic element and has multiple variable valences, so **X** is arsenic.
- B.** The molecular structure of **A** possesses a center of symmetry.
- C.** **B**, **C**, and **X** all have the same oxidation state for **X**, which is +4, so **X** is silicon.
- D.** **A** reacts with BCl_3 at low temperature to produce an ionic compound with a relative molecular mass of 345.98.
- E.** The valence of **X** in **F** and **G** are equal, both being +6.
- F.** The products of the reaction between **A** and O_3 are only **B** and **C**, with no other substances.
- G.** All of the above options are incorrect.

Answer

Correct Answer: **D**

Detailed Explanation

Element **X** is one of the essential non-metal elements for life, first narrowing the scope to C, N, O, P, S, Se, etc.

X_2Cl_2 is structurally similar to H_2O_2 , so **X** should be a chalcogen element.

A disproportionates upon hydrolysis to form an element and **B**, **B** should be XO_2 . It is also mentioned that it often exhibits oxidizing properties in chemical reactions. SO_2 mostly exhibits reducing properties, and SeO_2 is often used as an oxidant, so **X** is selenium.

CHECKPOINT

1 PTS

X is selenium

Therefore, the substances are as follows:

A: Se_2Cl_2

CHECKPOINT

1 PTS

A: Se_2Cl_2

B: SeO_2

CHECKPOINT

1 PTS

B: SeO_2

C: $SeCl_4$

CHECKPOINT

1 PTS

C: $SeCl_4$

D: SeF_4

CHECKPOINT

1 PTS

D: SeF_4

E: SeO_3

CHECKPOINT

1 PTS

E: SeO_3

F: SeO_2F_2

CHECKPOINT

1 PTS

F: SeO_2F_2

G: $SeOF_2$

CHECKPOINT

1 PTS

G: $SeOF_2$

H: H_2SeO_4

CHECKPOINT

1 PTS

H: H_2SeO_4

Based on this, analyze the options:

A, The element C is incorrect;

B is incorrect, **A** has no center of symmetry;

CHECKPOINT

1 PTS

A has no center of symmetry

E is incorrect, **G** has a valence of +4;

CHECKPOINT

1 PTS

G has a valence of +4

D is correct, **A** (Se_2Cl_2) reacts with BCl_3 at low temperature, which can be regarded as the transfer of Cl^- ions from Se_2Cl_2 to BCl_3 to generate $[Se_2Cl]^+[BCl_4]^-$ ionic compounds, and the relative molecular mass is correct;

F is incorrect, the reaction formula is $3Se_2Cl_2 + 4O_3 = 4SeO_2 + 2SeCl_4 + O_2$. In addition to **B** and **C**, oxygen is also produced;

CHECKPOINT

1 PTS

In addition to **B** and **C**, oxygen is also produced

G is incorrect because option D is correct.